

The Planck-Scale Structure of the Conversion Factors
Why $G = (\ell_P^2 \times c^3)/\hbar$ justifies the form of the factors from
Document 012 T0 Theory: From Dimensionless to SI

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Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Numerical verification, path from T0: $G_{\text{nat}} \times C_{\text{conv}} \times K_{\text{frak}}$ with $3.1 \times 10^{-10} \times 7.8 \times 10^{-3} \times 0.986 \times 10^1$ (gives 2.38×10^{-11}) $\rightarrow \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} \approx 6.69 \times 10^{-11}$, without C_{dim} .
- SI conversion factor: " $C_{\text{conv}} \sim c^3/\hbar$ (order of magnitude)" $\rightarrow$ structure only; numerically $c^3/\hbar = 2.6 \times 10^{59}$, and $C_{\text{conv}} = 7.8 \times 10^{-3}$ is the SI conversion of G .
- Code excerpt from calc_De.py: note, $G_{\text{nat}} \times 2.88294 \times 10^{-5} \approx 8.7 \times 10^{-15}$; $2.88294 \times 10^{-5} \neq C_{\text{conv}} K_{\text{frak}}$.
- Path via the Planck formula: checks the dimension, not the numerical value, since ℓ_P is defined via G (Doc. 012).
- G_{SI} formula: $\frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \times K_{\text{frak}} \rightarrow \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}}$ – with C_{dim} but without the compensating factor E_{char} , G would be too small by a factor of $E_{\text{char}} = 28.8$.
- Dimension of $c^3/\hbar$: $[G]/[\text{energy}] \rightarrow [G]/[\text{length}^2] = \text{m}/(\text{kg} \cdot \text{s}^2)$.
- Structure of the conversion factor $[E^{-2}] \rightarrow [\text{m}^3/(\text{kg} \cdot \text{s}^2)]: [\text{Length}^3]/[\text{Energy}] \rightarrow [\text{Length}^5 \cdot \text{Energy}]/[\text{Time}^4]$.

Abstract

This document explains why the conversion factors in Document 012 have the exact form they do. The mathematical relation $G = \frac{\ell_P^2 c^3}{\hbar}$ is not a new calculation method (it is a rearrangement of the known Planck length definition), but it reveals the *fundamental structure* underlying the conversion factors.

Core Message: The factors C_{dim} , C_{conv} , and K_{frak} in Document 012 are not arbitrary but follow from the Planck-scale structure of G . The formula also serves as a consistency check: if all factors are correct, $G_{\text{SI}} = \frac{\ell_P^2 c^3}{\hbar}$ must be satisfied.

For the complete technical derivation of all conversion factors, see Document 012.

1 The Problem: Conversion from T0 to SI

Recap: The T0 Formula for G

From Document 012 it is known:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (1)$$

(Corrected on 30 Sept. 2026: previously $\frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \times K_{\text{frak}} - C_{\text{dim}} = 1/E_{\text{char}} = 3.472 \times 10^{-2}$ is cancelled again by the factor $E_{\text{char}} = 28.8$ in the step chain of Doc. 012; in the formula alone the product would give 2.32×10^{-12} instead of 6.674×10^{-11} .)

With the factors:

- $\frac{\xi^2}{4m_e} \approx 8.7 \times 10^{-9} \text{ MeV}^{-1}$ (from T0 geometry)
- $C_{\text{dim}} \approx 3.5 \times 10^{-2}$ (dimension correction)

- $C_{\text{conv}} \approx 7.8 \times 10^{-3} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \cdot \text{MeV}$ (SI conversion)
- $K_{\text{frak}} = 0.986$ (fractal correction)

The Question

Why do these factors have exactly this form?

Specifically:

- Why does c^3 appear? (in C_{conv})
- Why $\hbar$ in the denominator?
- Why a length scale squared?
- What is the fundamental structure?

.2 The Planck Length as the Starting Point

Standard Definition (since Max Planck, 1899)

The Planck length is defined as:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}} \quad (2)$$

Standard Interpretation:

- G is a fundamental constant (measured)
- ℓ_P is calculated from it
- $\ell_P \approx 1.616 \times 10^{-35} \text{ m}$
- Quantum gravity scale

Mathematical Rearrangement

From $\ell_P = \sqrt{\frac{\hbar G}{c^3}}$ it follows by rearrangement:

$$\ell_P^2 = \frac{\hbar G}{c^3} \quad (3)$$

$$\ell_P^2 \times c^3 = \hbar G \quad (4)$$

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \quad (5)$$

This rearrangement is the structure to which the following sections refer.

.3 The Structure of the Conversion Factors

What Does the Planck Formula Show?

[Fundamental Structure]

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \quad (6)$$

Dimensional Analysis:

$$[G] = \frac{[\ell_P^2] \times [c^3]}{[\hbar]} \quad (7)$$

$$= \frac{[m^2] \times [m^3/s^3]}{[J \cdot s]} \quad (8)$$

$$= \frac{[m^5/s^3]}{[kg \cdot m^2/s^2 \cdot s]} \quad (9)$$

$$= \frac{[m^5/s^3]}{[kg \cdot m^2/s]} \quad (10)$$

$$= \frac{[m^3]}{[kg \cdot s^2]} \quad (11)$$

This yields the dimension $[G] = m^3/(kg \cdot s^2)$ (dimension check satisfied).

Connection to T0 Factors

In T0, one starts with G_{nat} in dimension $[E^{-2}]$ (Energy⁻²).

Conversion $[E^{-2}] \rightarrow [m^3/(kg \cdot s^2)]$ must have:

$$[E^{-2}] \times \text{Factor} = [m^3/(kg \cdot s^2)] \quad (12)$$

The factor must have the structure:

$$\text{Factor} = \frac{[\text{Length}^5 \cdot \text{Energy}]}{[\text{Time}^4]} \quad (13)$$

(Corrected on 30 Sept. 2026: previously $[\text{Length}^3]/[\text{Energy}]$ – the factor is $[G] \cdot [E^2] = m^3/(kg \cdot s^2) \cdot (kg \cdot m^2/s^2)^2 = kg \cdot m^7/s^6 = [\text{Length}^5 \cdot \text{Energy}]/[\text{Time}^4]$; $[\text{Length}^3]/[\text{Energy}] = m \cdot s^2/kg$ does not fit and contradicts the calculation below, which gives $[G] = [\text{Length}^5]/[\text{Energy} \cdot \text{Time}^4]$.)

From the Planck formula:

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \Rightarrow \text{Structure: } \frac{[\text{Length}^2] \times [\text{Velocity}^3]}{[\text{Action}]} \quad (14)$$

With $[\hbar] = [\text{Energy} \times \text{Time}]$ and $[c] = [\text{Length}/\text{Time}]$:

$$\frac{[\ell_P^2 \times c^3]}{[\hbar]} = \frac{[\text{Length}^2] \times [\text{Length}^3/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \quad (15)$$

$$= \frac{[\text{Length}^5/\text{Time}^3]}{[\text{Energy} \times \text{Time}]} \quad (16)$$

$$= \frac{[\text{Length}^5]}{[\text{Energy} \times \text{Time}^4]} \quad (17)$$

This justifies why:

- c^3 in the numerator ($\text{Length}^3/\text{Time}^3$)
- $\hbar$ in the denominator ($\text{Energy} \times \text{Time}$)
- Length^2 (from ℓ_P^2)
- The combination yields $[m^3/(kg \cdot s^2)]$

.4 Justification of the Factors in Document 012

The Dimension Correction Factor C_{dim}

From Document 012:

$$C_{\text{dim}} = \frac{1}{E_{\text{char}}} \approx 3.5 \times 10^{-2} \quad [\text{MeV}^{-1}] \quad (18)$$

With $E_{\text{char}} = 28.8 \text{ MeV}$ (7-step derivation in Doc. 012). *(Corrected on 30 Sept. 2026: previously 28.4 – the derivation in Doc. 012 gives 28.8, now uniform; E_{char} cancels in G.)*

Why this factor?

The T0 formula $G = \frac{\xi^2}{4m_e}$ initially yields dimension $[E^{-1}]$.

In natural units, however, G requires $[E^{-2}]$.

$\Rightarrow$ Factor $[E^{-1}]$ needed: $C_{\text{dim}} = 1/E_{\text{char}}$

Connection to the Planck structure:

The energy scale E_{char} is not arbitrary but emerges from the same geometry as ℓ_P . It is the characteristic scale where T0 geometry connects to the Planck scale.

The SI Conversion Factor C_{conv}

From Document 012:

$$C_{\text{conv}} \approx 7.8 \times 10^{-3} \quad [\text{m}^3 \text{kg}^{-1} \text{s}^{-2} \cdot \text{MeV}] \quad (19)$$

Structure of this factor:

$$C_{\text{conv}} \sim \frac{c^3}{\hbar} \quad (\text{in appropriate units}) \quad (20)$$

$$= \frac{(2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \approx 2.6 \times 10^{59} \quad (\text{SI numerical value, structure only}) \quad (21)$$

(Corrected on 30 Sept. 2026: previously “(order of magnitude)” – in SI, $c^3/\hbar = 2.6 \times 10^{59}$ does not correspond numerically to $C_{\text{conv}} = 7.8 \times 10^{-3}$; C_{conv} is the SI conversion of G , and the relation to $c^3/\hbar$ concerns only the structure.)

Why exactly this combination?

The Planck formula $G = \frac{\ell_P^2 c^3}{\hbar}$ shows:

- c^3 converts time scale to space scale (dimension: m^3/s^3)
- $\hbar$ connects energy with frequency (dimension: $\text{J} \cdot \text{s}$)
- Combination $c^3/\hbar$ has dimension $\text{m}/(\text{kg} \cdot \text{s}^2) = [G]/[\text{length}^2]$ *(Corrected on 30 Sept. 2026: previously “ $[\text{m}^3/(\text{kg} \cdot \text{s}^2)]/[\text{Energy}]$ ” – from $G = \ell_P^2 c^3/\hbar$ it follows that $[c^3/\hbar] = [G]/[\text{extlength}^2]$; $[G]/[E]$ would be m/kg^2 .)*

This is the role played by C_{conv} .

Numerical Verification

Consistency Check

From T0 (Document 012):

$$G_{\text{nat}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \approx 3.1 \times 10^{-10} \quad [E^{-2}] \quad (22)$$

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{conv}} \times K_{\text{frak}} \quad (23)$$

$$\approx 8.7 \times 10^{-9} \times 7.8 \times 10^{-3} \times 0.986 \quad (24)$$

$$\approx 6.69 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \quad (25)$$

(Corrected on 30 Sept. 2026: previously $G_{\text{SI}} = G_{\text{nat}} \times C_{\text{conv}} \times K_{\text{frak}} \approx 3.1 \times 10^{-10} \times 7.8 \times 10^{-3} \times 0.986 \times 10^1 \approx 6.67 \times 10^{-11}$ – this product gives 2.38×10^{-11} , and the factor 10^1 has no origin; only $\frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}}$ without C_{dim} is consistent.)

From Planck Formula (Verification):

$$\ell_P = 1.616 \times 10^{-35} \text{ m} \quad (26)$$

$$c = 2.998 \times 10^8 \text{ m/s} \quad (27)$$

$$\hbar = 1.055 \times 10^{-34} \text{ J} \cdot \text{s} \quad (28)$$

$$G_{\text{check}} = \frac{\ell_P^2 \times c^3}{\hbar} \quad (29)$$

$$= \frac{(1.616 \times 10^{-35})^2 \times (2.998 \times 10^8)^3}{1.055 \times 10^{-34}} \quad (30)$$

$$= \frac{2.611 \times 10^{-70} \times 2.694 \times 10^{25}}{1.055 \times 10^{-34}} \quad (31)$$

$$= 6.67 \times 10^{-11} \quad [\text{m}^3/(\text{kg} \cdot \text{s}^2)] \quad (32)$$

Both paths give $6.7 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$. The second path checks the dimension, not the numerical value, since ℓ_P is defined via G (see Doc. 012). (Corrected on 30 Sept. 2026: previously “Both paths give $6.67 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ to the digits shown.” – with the rounded factors the first path gives 6.69×10^{-11} ; the Planck formula only returns G because ℓ_P is computed from G .)

This shows: The factors in Doc. 012 have the dimensionally appropriate structure.

.5 The Role of the Planck Formula in T0

Not Circular in T0

Why is the formula not circular?

Standard Physics (circular):

1. Measure G
2. Calculate $\ell_P = \sqrt{\hbar G / c^3}$
3. Calculate $G = \ell_P^2 c^3 / \hbar$
 $\Rightarrow$ Get G back (no information gained)

T0 Physics (not circular):

1. Determine ξ from experiment (via α , E_0)

2. Calculate G_{SI} from ξ (with factors)
3. Calculate $\ell_P = \sqrt{\hbar G_{SI}/c^3}$
4. Check: $G_{SI} = \ell_P^2 c^3 / \hbar$
 $\Rightarrow$ Consistency check

Three Uses of the Planck Formula

1. **Justification:** Shows why factors have the form $c^3/\hbar$ etc.
2. **Verification:** Consistency check for calculated G
3. **Structural Insight:** G emerges at the Planck scale

.6 Practical Application: Python Implementation

Code Structure (from calc_De.py)

The T0 calculation script shows exactly this logic:

```
# Main calculation (from ξ)
G_t0_dimensionless = (xi**2) / (4 * m_char)
conversion_factor_nat = 3.47222e-2 # C_dim
G_nat = G_t0_dimensionless * conversion_factor_nat

SI_conversion_factor = 2.88294e-5 # C_conv × K_frak
G_SI = G_nat * SI_conversion_factor

# Planck formula as verification
planck_conversion_factor = (l_P**2 * c**3) / hbar

# Check: Both should agree
assert abs(G_SI - planck_conversion_factor) < 1e-13
```

(Note of 30 Sept. 2026: with the numbers of this code excerpt, $G_{nat} \times 2.88294 \times 10^{-5} \approx 8.7 \times 10^{-15}$, not G ; moreover, 2.88294×10^{-5} is not $C_{conv} \times K_{frak} \approx 7.7 \times 10^{-3}$. Consistent is $\frac{\xi^2}{4m_e} C_{conv} K_{frak}$ without C_{dim} , see the numerical verification.)

What the Code Shows

- **Lines 1-2:** T0 formula $\xi^2/(4m)$
- **Line 3:** Dimension correction C_{dim} (corresponds to $1/E_{char}$)
- **Line 5:** SI conversion $C_{conv} \times K_{frak}$ (corresponds to $c^3/\hbar$ structure)
- **Line 8:** Planck formula for verification
- **Line 11:** Both paths must agree.

.7 Comparison with Electrodynamics

Analogy: Speed of Light

In electrodynamics:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (33)$$

Interpretation:

- c emerges from electromagnetic vacuum structure
- μ_0, ϵ_0 describe vacuum properties
- Formula shows structure, not calculation

Analogy: Gravitational Constant

In T0:

$$G = \frac{\ell_P^2 \times c^3}{\hbar} \quad (34)$$

Interpretation:

- G emerges from spacetime geometry (T0)
- $\ell_P, c, \hbar$ describe geometry properties
- Formula shows structure, justifies conversion factors

Parallelism

Aspect	Electrodynamics	Gravitation
Constant	c	G
Formula	$c = 1/\sqrt{\mu_0 \epsilon_0}$	$G = \ell_P^2 c^3 / \hbar$
Emerges from	EM vacuum	Spacetime geometry
Justifies	μ_0, ϵ_0 structure	C_{conv} structure

Table 1: Parallel Structures

.8 Summary

The Central Message

Structure Justification

The Planck formula $G = \frac{\ell_P^2 \times c^3}{\hbar}$ is useful for T0 because it:

1. **Justifies** why the conversion factors in Doc. 012 have exactly the form:
 - $C_{\text{dim}} \sim 1/E$ (energy scale)
 - $C_{\text{conv}} \sim c^3/\hbar$ (Planck structure)
2. **Serves as a consistency check:**
 - Calculate G from ξ with factors
 - Calculate ℓ_P from G

- Check: $G = \ell_p^2 c^3 / \hbar$

3. Shows the geometric structure:

- G emerges at Planck scale ℓ_p
- Connection quantum mechanics ($\hbar$) $\leftrightarrow$ relativity (c)
- Fundamental role of geometry

It is not a new calculation method (that would be circular), but it justifies the structure of the factors.

What is New?

Mathematically NOT new:

- The formula $G = \ell_p^2 c^3 / \hbar$ (rearrangement of ℓ_p definition since 1899)
- The Planck units (Max Planck, 1899)

New in T0:

- The formula *justifies* the conversion factors
- It serves as *verification* (not circular, since G comes from ξ)
- It shows that G emerges at the Planck scale
- ℓ_p is not fundamental but follows from G (which follows from ξ)

Connection to Document 012

Document 012 shows: HOW to calculate G from ξ (all steps)

This document (127) shows: WHY the factors have this structure

Together: Calculation and structural justification of G in T0

Practical Significance

For calculations:

- Use the T0 path: $\xi \rightarrow G$ (Doc. 012)
- Planck formula as a check
- Both must agree

For understanding:

- Planck formula shows structure
- Justifies why $c^3/\hbar$ appears
- Shows geometric origin

For philosophy:

- G is not fundamental
- G emerges at the Planck scale
- G follows from ξ together with the conversion factors of Doc. 012